TECHNOLOGY

Comptia Security + 701
Course-end Project

Brute Force Attack

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Objectives

Conduct a comprehensive security assessment and response for a CentOS VM under brute force attack, focusing on log analysis, user verification, and implementation of enhanced security measures to mitigate future threats

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Prerequisites

- Linux System Administration & Scripting
- Incident Response
- Log Analysis & User Management
- Network Security & Monitoring

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Problem Statement and Motivation

Problem Statement:

Conduct a comprehensive security assessment and response for a CentOS VM under brute force attack, focusing on log analysis, user verification, and implementation of enhanced security measures to mitigate future threats

Real-World Scenario:

Your organization's critical CentOS virtual machine has been targeted by brute force, prompting an urgent need to analyze authentication logs. The IT security team will identify and check attempted usernames against the company's user database to determine if any valid accounts were targeted.

To counter the attack, the team will implement security enhancements, including changing the SSH port, enforcing stricter password policies, enabling multifactor authentication, and setting up continuous monitoring. These actions aim to stop the current threat and bolster defenses against future attacks.

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Industry Relevance

Skills used in the project and their usage in the industry are given below:

1. **Log Analysis & Scripting Tools:** Tools like grep, awk, and scripting languages such as Bash or Python for log analysis and automation.

2. **Network Security & Access Tools:** Tools like iptables, fail2ban, and SSH clients (e.g., OpenSSH, PuTTY) for securing access and configuring firewalls.

3. **Multifactor Authentication (MFA) & User Management Tools:** Tools like Google Authenticator or Duo Security for MFA, along with command-line utilities for user account management.Continuous

4. **Monitoring & Documentation Tools:** Tools like Nagios or Zabbix for real-time monitoring, and Git or Markdown editors for version control and documentation.

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Task (Activities)

1. **Download Authentication Logs:** Access and download authentication logs from the provided URL. These logs contain critical evidence of brute force attacks, including access attempts and usernames.

2. **Analyze the Logs for Usernames:** Use log analysis tools or scripts to extract all usernames attempted during the attack, identifying the extent and specific entry points targeted.

3. **Cross-Reference Usernames with Company Records:** Cross-reference extracted usernames with the internal user database to check if any correspond to actual user accounts, indicating potential insider threats.

4. **Implement Security Enhancements:** Based on findings, enhance security by enforcing stricter password policies, implementing multifactor authentication, and possibly changing SSH ports.

5. **Continuous Monitoring and Reporting:** Set up continuous monitoring to detect unusual access patterns and generate regular reports to inform the security team of any new threats.

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Submission Process

You will have to submit the project in one week.

It is recommended to work in the integrated labs, as they have all the required tools available.

The project can be submitted from the assessment tab by clicking the **Submit** button.

Provide the documents mentioned below:

- Source Code in zip
- Database scripts to replicate your database settings
- Screenshots of the outputs

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Thank You

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